DEVICEOPS / PROFESSIONAL AGENTS

Autonomous Operations & Incident Resolution

Observe → investigate → authorize → act → verify. Multi-agent reasoning on AWS Bedrock AgentCore & Alibaba Qwen.

01 DEVICE & DELIVERY LAYER

Controlled LAN / local notification services

OBSERVATION SOURCES

Devices & simulators

LAN equipment: read-only probes
Standalone HTTP simulator
Physical/cloud writes disabled

NODE.JS EDGE COLLECTOR

Normalize & deliver

Ping · SNMP system / ports · PJLink
Encrypted vault + durable spool
Outbound API + command journal

MOBILE PHOTO CAPTURE

Authenticated upload to v2 API

SHA-256 + incident association

INTEGRATION BOUNDARY

57-entry connector catalog

12 mock-tested adapters; 45 planned.
Physical hardware unverified.

LOCAL NOTIFICATIONS

n8n → Mailpit

Email delivery + signed callback

telemetry

02 DEVICEOPS APPLICATION

Next.js + TypeScript worker + PostgreSQL

OPERATOR ENTRY POINT

Next.js operations UI + v2 API

Sessions · tenant scope · manager approvals
Collector ingestion · MCP tool endpoint

persisted state

AUTONOMOUS ORCHESTRATION

TypeScript operations worker

EWMA drift: temp, memory, optical loss
PostgreSQL-backed pg-boss investigation jobs
Campus topology & calculated impact

validated proposal

BACKEND OWNS AUTHORIZATION

Policy → execution → verification

Service restart: automatic when policy allows
Switch restart: manager approval + exact hash
One command; 3 fresh healthy samples or escalate

PERSISTENCE

PostgreSQL 17

Tenant RLS · baselines · memory · jobs · audit

investigate

03 AWS AGENTCORE & REASONING

Bedrock AgentCore + Alibaba Cloud Qwen

ACTIVE REASONING PROVIDER

Alibaba Cloud Model Studio

qwen3.7-max in recorded recovery
OpenAI-compatible model inference
Explicit model selection · cited findings

inference

SERVERLESS AGENT RUNTIME

Amazon Bedrock AgentCore

Strands SDK · bounded specialist workflow
Network / System / Power → Critic
Python 3.12 ARM64 microVM in AWS us-east-1
Direct S3 code bundle · HTTP invocations
Proposes actions; backend owns authorization

tool calls

RESTRICTED HYBRID TOOL BRIDGE

Cloudflare Tunnel → Tool Proxy

Grant checks enforced by local backend
Ephemeral HTTPS/QUIC tunnel to loopback :3035
Allowed tool + invocation-status callbacks
DB stays local; evidence goes to cloud

RETRIEVAL Local lexical runbook search; optional Titan embeddings. The recorded cloud recovery does not establish live vector retrieval.

END-TO-END INCIDENT LIFECYCLE

Observe → Detect & group → Investigate → Authorize → Execute → Verify → Report / notify

MULTIMODAL EVIDENCE

Quarantine → scan → isolated parser

Mobile photo & manual PDF (>pg 5 citations)
ClamAV malware scan + SHA-256 integrity
Device associations require confirmation
Only clean, approved artifacts enter context

SANDBOXED ANALYSIS

Private broker → disposable container

No network · credentials · application mounts
1 CPU · 256 MiB · 64 PIDs · 30 seconds
Telemetry CSV + synthetic waveform FFT
Measured spectra; no bearing-fault diagnosis

ORGANIZATIONAL MEMORY

Incident / knowledge / fleet memory

Factual outcomes + reviewed runbooks
Source IDs, versions and tenant isolation
EWMA baselines + explicit evidence gaps
Lexical retrieval; optional Titan embeddings

RECORDED VERIFICATION · 14 SEP 2026

AgentCore + Qwen: autonomous service recovery verified in 100.5s; authenticated manager approval & switch recovery verified in 114.3s.